Practice workbook · Translators: assembler, compiler and interpreter

What each translator does, how compilers and interpreters differ, their advantages, and choosing a translator for a task.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-16

1. Question 5 · part a · 3 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part a only.

5 A team is developing a computer game.

(a) Complete the table by adding a tick (✓) to show the characteristics of program language translators.

You may select more than one translator per characteristic.

(3)

Characteristic	Translator		
	Assembler	**Interpreter**	**Compiler**
Does **not** have to be present in memory to execute the translated code.			
Translates high level language source code.			
Executes high level code until either the program has finished, or a syntax error is encountered.			
Translates low level language source code into machine code.			

(b) Luca is testing the computer game.

The computer that he is using is an older model with a processor that runs at a low clock speed.

Explain **one** drawback of testing the game using this computer.

(2)

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P 7 5 7 3 6 A 0 1 8 2 8

1. Question 5 · part a · 3 marks · Working and answer

Working and answer

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2. Question 6 · part c · 6 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

6 A software engineer is working with some algorithms.

(a) An algorithm needs to be developed.

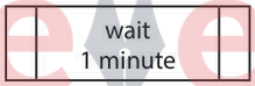
The algorithm must:

- check for a change of hour every minute
- use the library subprogram getHour() to get the hour part of the current time in the 24-hour clock (0 to 23)
- output "Good morning" when the hour is between 3 and 12, inclusive
- output "Good afternoon" when the hour is between 13 and 19, inclusive
- output "Good night" at all other times.

Complete the flowchart to represent this algorithm, in the space provided on the next page.

The call to the library subprogram getHour() has already been included.

Use this subprogram symbol to show waiting for 1 minute.



(6)

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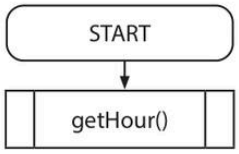
2. Question 6 · part c · 6 marks

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Further practice · Attempt part c only.

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Draw your flowchart here.



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2. Question 6 · part c · 6 marks

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Further practice · Attempt part c only.

(b) Another algorithm determines when to change the flavoured syrups for a drinks dispensing machine.

Figure 4 shows this algorithm written in pseudocode.

```
 1 SET flavours TO ["anise", "mango", "cola", "apple", "papaya", "strawberry",
   "lychee", "banana"]
 2
 3 SET volume TO [0.7, 0.2, 0.6, 0.1, 0.05, 0.8, 0.4, 0.6]
 4
 5 SET i TO 0
 6
 7 WHILE i < 8 DO
 8
 9     IF (volume[i] < 0.1) THEN
10         SEND (flavours[i] & " needs changing") TO DISPLAY
11
12     ELSE
13         IF (volume[i] >= 0.3) AND (volume[i] <= 0.5) THEN
14             SEND (flavours[i] & " needs ordering")  TO DISPLAY
15         END IF
16     END IF
17
18     SET i TO i + 1
19
20 END WHILE
```

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Figure 4

(i) Give the output produced by the algorithm.

(2)

The algorithm works with a fixed number of flavours.

(ii) Give the number of the line in the algorithm that would need to be amended to allow for any number of flavours.

(1)

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2. Question 6 · part c · 6 marks

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Further practice · Attempt part c only.

(iii) State how the pseudocode needs to be changed to make this amendment.

(1)

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QUESTION 6(c) BEGINS ON THE NEXT PAGE.

2. Question 6 · part c · 6 marks

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Further practice · Attempt part c only.

(c) The algorithms will be coded in a high-level language.

Programs written in high-level languages must be translated in order to be executed by a computer.

Discuss the similarities and differences between a compiler and an interpreter.

(6)

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(Total for Question 6 = 16 marks)

2. Question 6 · part c · 6 marks · Working and answer

Working and answer

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3. Question 5 · part c · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

5 Isaac is a program developer.

(a) **Figure 3** shows an algorithm Isaac has written.

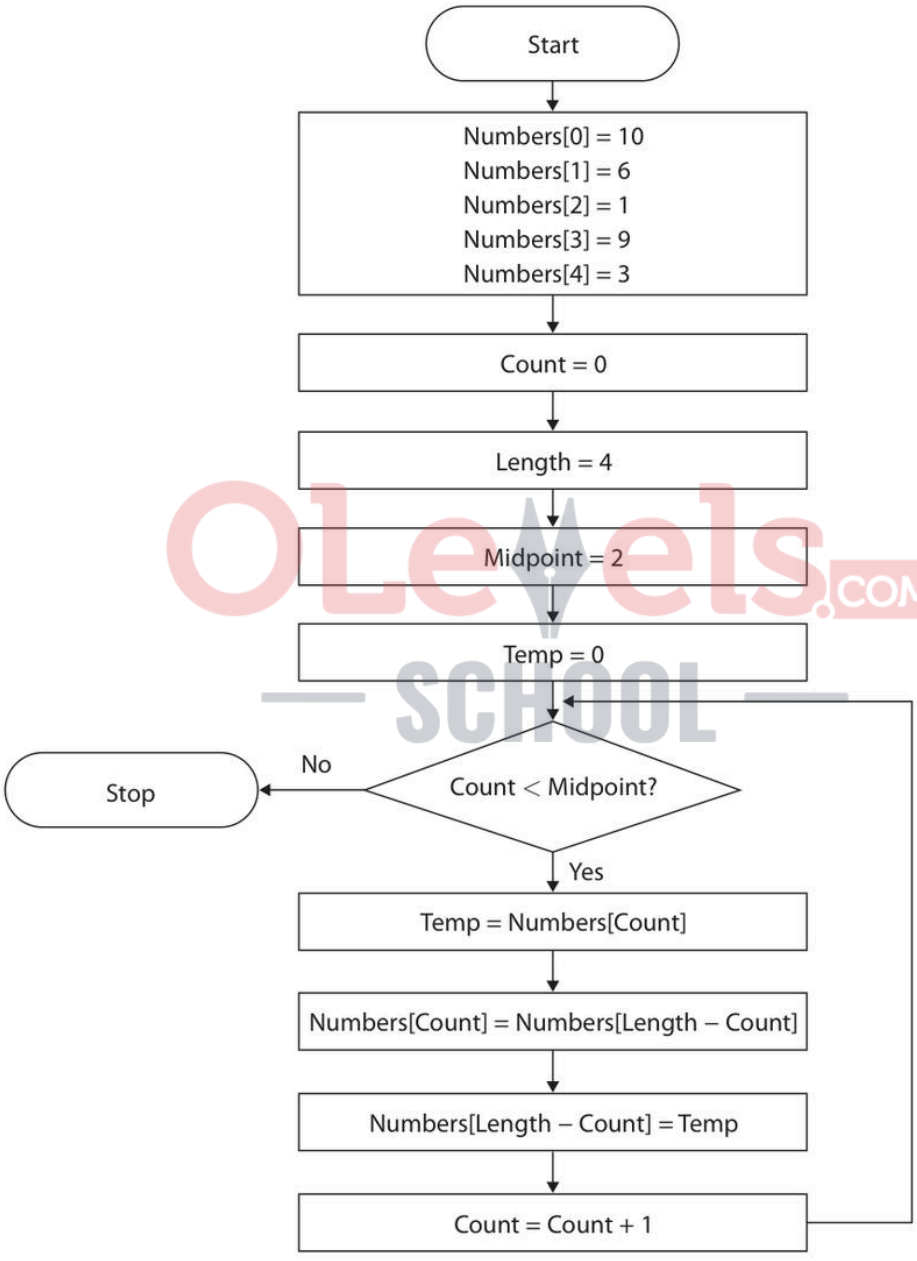


Figure 3

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P 6 6 4 8 9 A 0 1 2 2 0

3. Question 5 · part c · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

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(i) Complete the trace table. You may not need to use all of the rows.

(5)

				Numbers array				
Count	**Length**	**Midpoint**	**Temp**	[0]	[1]	[2]	[3]	[4]
0	4	2	0	10	6	1	9	3

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(ii) Give the purpose of the algorithm.

(1)

.....

.....

(iii) Explain why the variable Temp is needed.

(2)

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3. Question 5 · part c · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

(b) **Figure 4** shows an algorithm Isaac has written using pseudocode.

The algorithm should display the average of the numbers that have been input.

```
1    SET total TO 0
2    SET number TO 0
3    SET count TO 0
4    WHILE number <> –1 DO
5       SEND 'Input a number or –1 to end the program' TO DISPLAY
6       RECEIVE number FROM (INTEGER) KEYBOARD
7       SET total TO total + number
8       SET count TO count + 1
9    END WHILE
10   SET average TO total / count
11   SEND 'The average is ' & average TO DISPLAY
```

Figure 4

Isaac uses the input 2, 3, 5, 2, –1 to test the algorithm. He discovers an error.

Expected result	Actual result
The average is 2.75	The average is 2.2

(i) Explain why the **Actual** result is not the same as the **Expected** result.

(2)

(ii) Give the number of the line that contains the error.

(1)

(iii) Amend a single line of pseudocode to correct the error.

(1)

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P 6 6 4 8 9 A 0 1 4 2 0

3. Question 5 · part c · 2 marks

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Further practice · Attempt part c only.

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(c) Isaac is going to sell one of his programs and distribute it on a DVD.

Give **two** advantages of choosing a programming language that uses a compiler, rather than an interpreter, for a program that will be distributed on a DVD.

(2)

1 .....

.....

2 .....

.....

(Total for Question 5 = 14 marks)

Working and answer

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4. Question 1 · part e.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part e.ii only.

1 Computer systems have both hardware and software components.

(a) The central processing unit (CPU) uses the fetch-decode-execute cycle.

(i) State what is meant by the term **program instruction**.

(1)

(ii) State what is meant by the term **memory address**.

(1)

(b) Identify the component of the CPU that provides temporary data storage.

(1)

- ☒ **A** Address bus
- ☒ **B** Data bus
- ☒ **C** Control unit
- ☒ **D** Register

(c) The performance of the CPU is affected by the clock speed.

(i) Give **one** benefit of having a higher clock speed.

(1)

(ii) Give **one** drawback of having a higher clock speed.

(1)

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4. Question 1 · part e.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part e.ii only.

(d) Identify which **one** of these describes a sequential computational model.

(1)

- ☒ **A** Program instructions are read one after another from external storage
- ☒ **B** Program instructions are executed by multiple agents working together
- ☒ **C** Program instructions are executed in parallel by different cores
- ☒ **D** Program instructions are executed one after another

(e) A program can be written in a high-level or a low-level language.

(i) Give **one** reason for writing a program in a low-level language.

(1)

.....

.....

(ii) State the purpose of an assembler.

(1)

.....

(iii) Complete the table by adding **one** tick (✓) in **each** row to match the description.

(3)

Description	Compiler	Interpreter
Translates the program each time it is executed		
Produces permanent object code		
Translates line by line		
Translates the whole program before it is run		
Generates a list of errors once the complete program has been translated		

(Total for Question 1 = 11 marks)

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4. Question 1 · part e.ii · 1 mark · Working and answer

Working and answer

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5. Question 1 · part e.iii · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part e.iii only.

1 Computer systems have both hardware and software components.

(a) The central processing unit (CPU) uses the fetch-decode-execute cycle.

(i) State what is meant by the term **program instruction**.

(1)

(ii) State what is meant by the term **memory address**.

(1)

(b) Identify the component of the CPU that provides temporary data storage.

(1)

- ☒ **A** Address bus
- ☒ **B** Data bus
- ☒ **C** Control unit
- ☒ **D** Register

(c) The performance of the CPU is affected by the clock speed.

(i) Give **one** benefit of having a higher clock speed.

(1)

(ii) Give **one** drawback of having a higher clock speed.

(1)

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5. Question 1 · part e.iii · 3 marks

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Further practice · Attempt part e.iii only.

(d) Identify which **one** of these describes a sequential computational model.

(1)

- ☒ **A** Program instructions are read one after another from external storage
- ☒ **B** Program instructions are executed by multiple agents working together
- ☒ **C** Program instructions are executed in parallel by different cores
- ☒ **D** Program instructions are executed one after another

(e) A program can be written in a high-level or a low-level language.

(i) Give **one** reason for writing a program in a low-level language.

(1)

.....

.....

(ii) State the purpose of an assembler.

(1)

.....

(iii) Complete the table by adding **one** tick (✓) in **each** row to match the description.

(3)

Description	Compiler	Interpreter
Translates the program each time it is executed		
Produces permanent object code		
Translates line by line		
Translates the whole program before it is run		
Generates a list of errors once the complete program has been translated		

(Total for Question 1 = 11 marks)

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5. Question 1 · part e.iii · 3 marks · Working and answer

Working and answer

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6. Question 3 · part f · 4 marks

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Further practice · Attempt part f only.

3 Software programs carry out tasks on computers.

(a) A headteacher uses a computer simulation to model the effect of increasing class sizes.

Explain **one** problem with using simulations to predict the effects of changes.

(2)

(b) Images are stored as files before being displayed or printed.

(i) Give **one** measurement of image resolution.

(1)

(ii) Increasing the colour depth of an image leads to an increase in the image file size.

Describe the reason for the increase in file size.

(2)

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6. Question 3 · part f · 4 marks

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Further practice · Attempt part f only.

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(c) Software is divided into two categories.

Describe **one** difference between system software and application software.

(2)

(d) Here is an image of secondary storage.

Two files (W and Z) are stored on it.

Each file is made up of several blocks (e.g. Z1, Z2, Z3).

	W4	Z1				W2		W3	Z3		W1		Z2

Complete the image to show the state after running a defragmentation utility.

(2)

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6. Question 3 · part f · 4 marks

2023 JUN · Paper 01

Further practice · Attempt part f only.

(e) Here is part of a file that contains electric meter readings.

04631 04984 05103 05163 05271 05383 05487 05722

Explain the effect on the file of applying a run-length encoding algorithm to this data file.

(2)

.....

.....

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.....

(f) A series of tasks is written in different programming languages.

Complete the table to show the correct programming language translator for each task.

(4)

Task	Translator
A guessing game that can be used on different computing platforms	
A screen driver for a new smartphone	
A new version of a spreadsheet program for sale next year	
Control software for an embedded system inside a new washing machine	

(Total for Question 3 = 15 marks)

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6. Question 3 · part f · 4 marks · Working and answer

Working and answer

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7. Question 5 · part b.i · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b.i only.

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5 Smartwatches are a form of wearable technology.

(a) Embedded systems are used in smartwatches.

(i) A smartwatch has a step tracker that uses a built-in accelerometer sensor.

Complete the table by adding **one other** built-in sensor a smartwatch could use and its purpose.

(2)

Sensor	Purpose
Accelerometer	To measure body movement and step count

(ii) One characteristic of an embedded system in a smartwatch is that it uses sensors.

Give **two other** characteristics of an embedded system that make it suitable for use in a smartwatch.

(2)

1 .....

.....

2 .....

.....

(b) A software development team writes the software for a smartwatch using assembly language.

(i) Identify the purpose of an assembler.

(1)

- ☒ **A** To convert a low-level language into a high-level language
- ☒ **B** To convert low-level source code into machine code
- ☒ **C** To execute machine code
- ☒ **D** To translate a high-level language into a low-level language

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7. Question 5 · part b.i · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b.i only.

(ii) Explain **one** benefit of writing the software for the smartwatch in a low-level language.

(2)

(iii) Members of the software development team are computing professionals.

Give **two** ways that members of the team can demonstrate ethical behaviour.

(2)

1 .....

2 .....

(iv) The software for the smartwatch takes up 500,000 bytes of storage.

Identify the value that is equivalent to 500,000 bytes.

(1)

- ☒ **A** 0.5 MB
- ☒ **B** 500 KiB
- ☒ **C** 0.5 MiB
- ☒ **D** 5 MB

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7. Question 5 · part b.i · 1 mark · Working and answer

Working and answer

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